

# Regression Modelling

# Regression

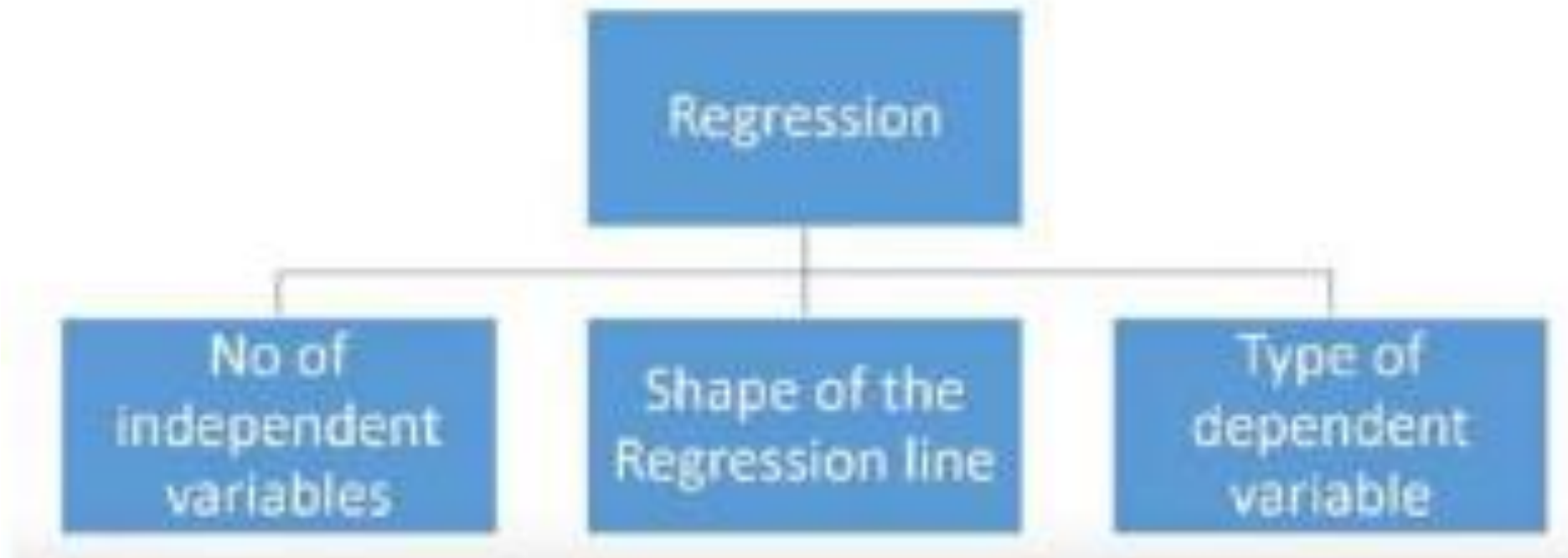
- Find different variables are related to one another.
- Basic idea
  1. **“Use the existing historical data to identify potential relationships among variables”**
  2. **“use these relationships to make predictions about the future”**

# Regression analysis

- Statistical process for estimating the relationships among variables
- It helps to model and analyse several variables when the focus is on the relationship between a dependent variable and one or more independent variables (or “predictors”)
  - Eg Petrol prices increase with increase in demand
- Study how SET OF PREDICTORS affects the behaviour of the dependent variable and allows us to use numerical values to model these effects.
- Regression analysis is widely used for **prediction and forecasting**

# Type of Regression Techniques

- These techniques are mostly driven by three metrics

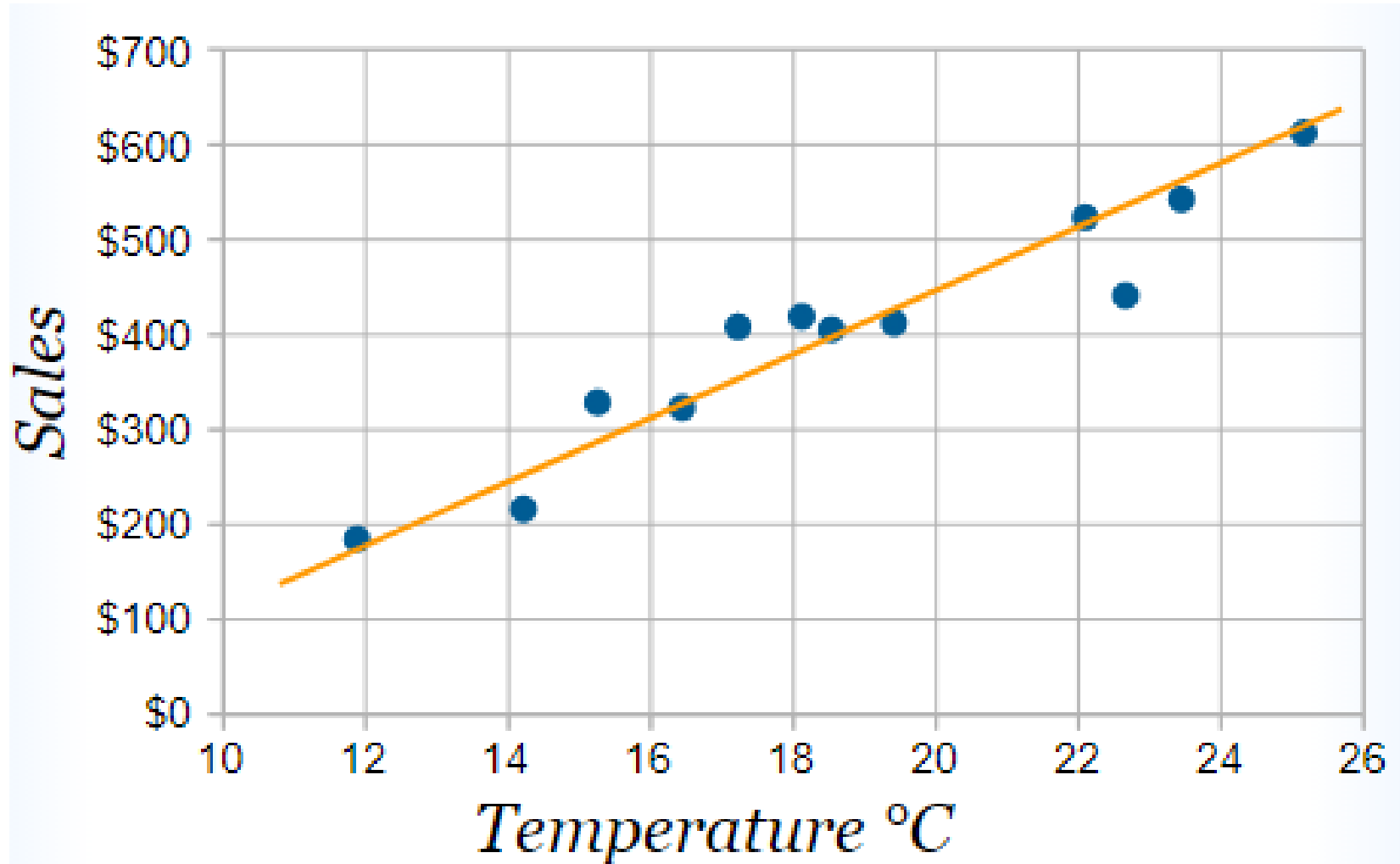


# Use of Regression Analysis

- There are multiple benefits of using regression analysis. They are as follows:
  1. It indicates the significant relationships between dependent variable and independent variable.
  2. It indicates the strength of impact of multiple independent variables on a dependent variable.
- Help to compare the effects of variables measured on different scales
  - Effect of price changes and the number of promotional activities.
  - Benefits market researchers / data analysts / data scientists to eliminate and evaluate the best set of variables to be used for building predictive models.

# Linear Regression

- **Line of Best Fit**
- Imagine you have some points, and want to have a **line** that best fits them like this
- But for better accuracy we will calculate the line using **Least Squares Regression**.



# Least Squares Regression.

- Our aim is to calculate the values  $m$  (slope) and  $b$  (y-intercept) in the equation of a line :

$$y = mx + b$$

- Where:
- **y** = how far up
- **x** = how far along
- **m** = Slope or Gradient (how steep the line is)
- **b** = the Y Intercept (where the line crosses the Y axis)

# To find the line of best fit for **N** points:

**Step 1:** For each (x,y) point calculate  $x^2$  and  $xy$

**Step 2:** Sum all  $x$ ,  $y$ ,  $x^2$  and  $xy$ , which gives us  $\Sigma x$ ,  $\Sigma y$ ,  $\Sigma x^2$  and  $\Sigma xy$  ( $\Sigma$  means "sum up")

**Step 3:** Calculate Slope **m**:

$$m = \frac{N \Sigma(xy) - \Sigma x \Sigma y}{N \Sigma(x^2) - (\Sigma x)^2}$$

(N is the number of points.)

**Step 4:** Calculate Intercept **b**:

$$b = \frac{\Sigma y - m \Sigma x}{N}$$

**Step 5:** Assemble the equation of a line

$$y = mx + b$$

Done!

- Sam found how many **hours of sunshine** vs how many **ice creams** were sold at the shop from Monday to Friday:

| "x"<br>Hours of<br>Sunshine | "y"<br>Ice Creams<br>Sold |
|-----------------------------|---------------------------|
| 2                           | 4                         |
| 3                           | 5                         |
| 5                           | 7                         |
| 7                           | 10                        |
| 9                           | 15                        |
| <b>8</b>                    | <b>???</b>                |

- Let us find the best **m** (slope) and **b** (y-intercept) that suits that data

$$y = mx + b$$

- $m = \frac{N \sum(xy) - \sum x \sum y}{N \sum(x^2) - (\sum x)^2}$

$$b = \frac{\sum y - m \sum x}{N}$$

- **Step 1:** For each (x,y) calculate  $x^2$  and  $xy$ :

| x | y  | $x^2$ | xy  |
|---|----|-------|-----|
| 2 | 4  | 4     | 8   |
| 3 | 5  | 9     | 15  |
| 5 | 7  | 25    | 35  |
| 7 | 10 | 49    | 70  |
| 9 | 15 | 81    | 135 |

- **Step 2:** Sum  $x$ ,  $y$ ,  $x^2$  and  $xy$  (gives us  $\Sigma x$ ,  $\Sigma y$ ,  $\Sigma x^2$  and  $\Sigma xy$ ):

| $x$                              | $y$                              | $x^2$                               | $xy$                               |
|----------------------------------|----------------------------------|-------------------------------------|------------------------------------|
| 2                                | 4                                | 4                                   | 8                                  |
| 3                                | 5                                | 9                                   | 15                                 |
| 5                                | 7                                | 25                                  | 35                                 |
| 7                                | 10                               | 49                                  | 70                                 |
| 9                                | 15                               | 81                                  | 135                                |
| <b><math>\Sigma x</math>: 26</b> | <b><math>\Sigma y</math>: 41</b> | <b><math>\Sigma x^2</math>: 168</b> | <b><math>\Sigma xy</math>: 263</b> |

- **Step 3:** Calculate Slope **m**:

- $$m = \frac{N \sum(xy) - \sum x \sum y}{N \sum(x^2) - (\sum x)^2}$$

- $$m = \frac{5 \times 263 - 26 \times 41}{5 \times 168 - 26^2}$$

- $$m = \frac{1315 - 1066}{840 - 676}$$

- $$m = 1.5183...$$

| x             | y             | x <sup>2</sup>             | xy              |
|---------------|---------------|----------------------------|-----------------|
| 2             | 4             | 4                          | 8               |
| 3             | 5             | 9                          | 15              |
| 5             | 7             | 25                         | 35              |
| 7             | 10            | 49                         | 70              |
| 9             | 15            | 81                         | 135             |
| <b>Σx: 26</b> | <b>Σy: 41</b> | <b>Σx<sup>2</sup>: 168</b> | <b>Σxy: 263</b> |

- **Step 4:** Calculate Intercept **b**:

- $$b = \frac{\Sigma y - m \Sigma x}{N}$$

- $$b = \frac{41 - 1.5183 \times 26}{5}$$

- $$b = 0.3049$$

- **Step 5:** Assemble the equation of a line:

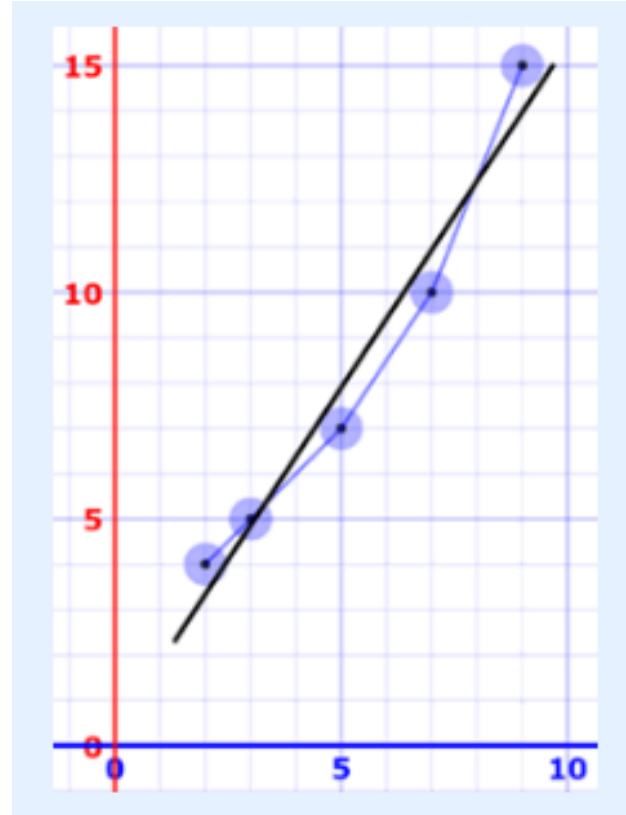
- $$y = mx + b$$

- $$y = 1.518x + 0.305$$

- **Step 5:** Assemble the equation of a line:
- $y = mx + b$
- $y = 1.518x + 0.305$

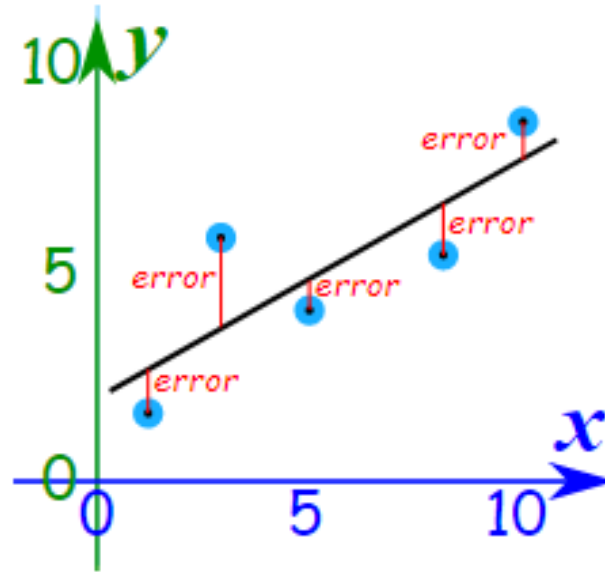
| x | y  | $y = 1.518x + 0.305$ | error |
|---|----|----------------------|-------|
| 2 | 4  | 3.34                 | -0.66 |
| 3 | 5  | 4.86                 | -0.14 |
| 5 | 7  | 7.89                 | 0.89  |
| 7 | 10 | 10.93                | 0.93  |
| 9 | 15 | 13.97                | -1.03 |

- Here are the (x,y) points and the line  $y = 1.518x + 0.305$  on a graph:



- For 8 hours of Sun how many ice-creams Sam will sell?
- $y = 1.518 \times 8 + 0.305 = 12.45$  Ice Creams

- It works by making the total of the **square of the errors** as small as possible (that is why it is called "least squares"):



The straight line minimizes the sum of squared errors

- So, when we square each of those errors and add them all up, the total is as small as possible.
- Least squares is sensitive to outliers. A strange value will pull the line towards it.

- a) Find the least square regression line for the following set of data

$$\{(-1, 0), (0, 2), (1, 4), (2, 5)\}$$

- Also Plot the given points and the regression line in the same rectangular system of axes.

- Create the table first

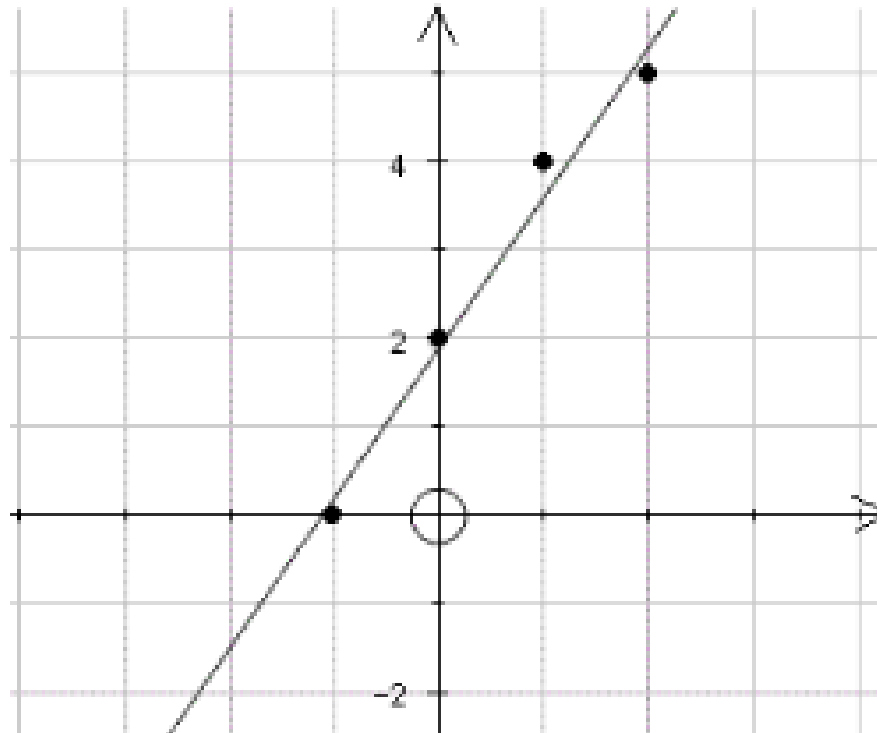
| <b>x</b>       | <b>y</b>        | <b>x y</b>        | <b>x<sup>2</sup></b> |
|----------------|-----------------|-------------------|----------------------|
| -1             | 0               | 0                 | 1                    |
| 0              | 2               | 0                 | 0                    |
| 1              | 4               | 4                 | 1                    |
| 2              | 5               | 10                | 4                    |
| $\Sigma x = 2$ | $\Sigma y = 11$ | $\Sigma x y = 14$ | $\Sigma x^2 = 6$     |

- We now use the above formula to calculate a and b as follows  

$$m = (n\sum x y - \sum x \sum y) / (n\sum x^2 - (\sum x)^2) = (4*14 - 2*11) / (4*6 - 2^2) = 17/10 = 1.7$$

$$b = (1/n)(\sum y - a \sum x) = (1/4)(11 - 1.7*2) = 1.9$$

- We now graph the regression line given by  $y = ax + b$  and the given points.



# Solve

- The sales of a company (in million dollars) for each year are shown in the table below.

|              |      |      |      |      |      |
|--------------|------|------|------|------|------|
| x<br>(year)  | 2005 | 2006 | 2007 | 2008 | 2009 |
| y<br>(sales) | 12   | 19   | 29   | 37   | 45   |

- We first change the variable  $x$  into  $t$  such that  $t = x - 2005$  and therefore  $t$  represents the number of years after 2005. Using  $t$  instead of  $x$  makes the numbers smaller and therefore manageable. The table of values becomes.

|                             |           |           |           |           |           |
|-----------------------------|-----------|-----------|-----------|-----------|-----------|
| <b>t (years after 2005)</b> | <b>0</b>  | <b>1</b>  | <b>2</b>  | <b>3</b>  | <b>4</b>  |
| <b>y (sales)</b>            | <b>12</b> | <b>19</b> | <b>29</b> | <b>37</b> | <b>45</b> |

- We now use the table to calculate  $a$  and  $b$  included in the least regression line formula.

- Create the table

| <b>t</b>        | <b>y</b>         | <b>t y</b>        | <b>t<sup>2</sup></b> |
|-----------------|------------------|-------------------|----------------------|
| 0               | 12               | 0                 | 0                    |
| 1               | 19               | 19                | 1                    |
| 2               | 29               | 58                | 4                    |
| 3               | 37               | 111               | 9                    |
| 4               | 45               | 180               | 16                   |
| $\Sigma t = 10$ | $\Sigma y = 142$ | $\Sigma ty = 368$ | $\Sigma t^2 = 30$    |

- We now calculate m and b using the least square regression formulas for a and b.

$$m = (n\sum t y - \sum t \sum y) / (n\sum t^2 - (\sum t)^2) = (5*368 - 10*142) / (5*30 - 10^2) = 8.4$$

$$b = (1/n)(\sum y - a \sum x) = (1/5)(142 - 8.4*10) = 11.6$$

- In 2012,  $t = 2012 - 2005 = 7$

The estimated sales in 2012 are:  $y = 8.4 * 7 + 11.6 = 70.4$  million dollars.

# Multiple Linear Regression

# REGIONAL DELIVERY SERVICE

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Let's assume that you are a small business owner for Regional Delivery Service, Inc. (RDS) who offers same-day delivery for letters, packages, and other small cargo. You are able to use Google Maps to group individual deliveries into one trip to reduce time and fuel costs. Therefore some trips will have more than one delivery.

As the owner, you would like to be able to *estimate how long a delivery will take* based on two factors: 1) the total distance of the trip in miles and 2) the number of deliveries that must be made during the trip.

# RDS DATA AND VARIABLE NAMING

To conduct your analysis you take a random sample of 10 past trips and record three pieces of information for each trip: 1) total miles traveled, 2) number of deliveries, and 3) total travel time in hours.

| milesTraveled,<br>( $x_1$ ) | numDeliveries,<br>( $x_2$ ) | travelTime(hrs),<br>( $y$ ) |
|-----------------------------|-----------------------------|-----------------------------|
| 89                          | 4                           | 7                           |
| 66                          | 1                           | 5.4                         |
| 78                          | 3                           | 6.6                         |
| 111                         | 6                           | 7.4                         |
| 44                          | 1                           | 4.8                         |
| 77                          | 3                           | 6.4                         |
| 80                          | 3                           | 7                           |
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| 109                         | 5                           | 7.3                         |
| 76                          | 3                           | 6.4                         |

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| 111                         | 6                           | 7.4                         |
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| 77                          | 3                           | 6.4                         |
| 80                          | 3                           | 7                           |
| 66                          | 2                           | 5.6                         |
| 109                         | 5                           | 7.3                         |
| 76                          | 3                           | 6.4                         |

Remember that in this case, you would like to be able to **predict the total travel time** using both the miles traveled and number of deliveries on each trip.

In what way does travel time **DEPEND** on the first two measures?

Travel time is the *dependent variable* and miles traveled and number of deliveries are independent variables.

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| 66                          | 1                           | 5.4                         |
| 78                          | 3                           | 6.6                         |
| 111                         | 6                           | 7.4                         |
| 44                          | 1                           | 4.8                         |
| 77                          | 3                           | 6.4                         |
| 80                          | 3                           | 7                           |
| 66                          | 2                           | 5.6                         |
| 109                         | 5                           | 7.3                         |
| 76                          | 3                           | 6.4                         |

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In what way does travel time **DEPEND** on the first two measures?

Travel time is the *dependent variable* and miles traveled and number of deliveries are independent variables.

Note: Some prefer *predictor variable(s)* and *response variable* instead of independent and dependent variables, respectively.

# MULTIPLE REGRESSION

Multiple regression is an extension of simple linear regression.

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Simple linear regression  
*one-to-one*

# MULTIPLE REGRESSION

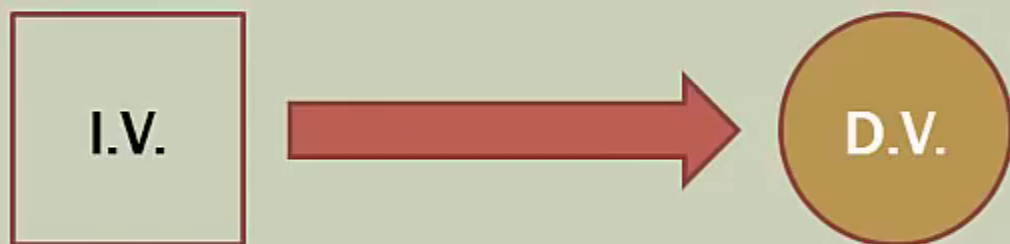
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Simple linear regression  
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# MULTIPLE REGRESSION

Multiple regression is an extension of simple linear regression.



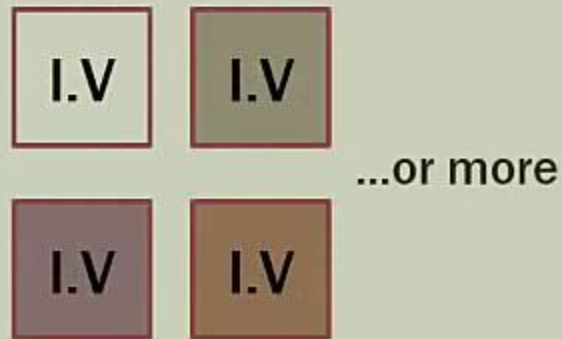
Simple linear regression  
*one-to-one*

# MULTIPLE REGRESSION

Multiple regression is an extension of simple linear regression.



Simple linear regression  
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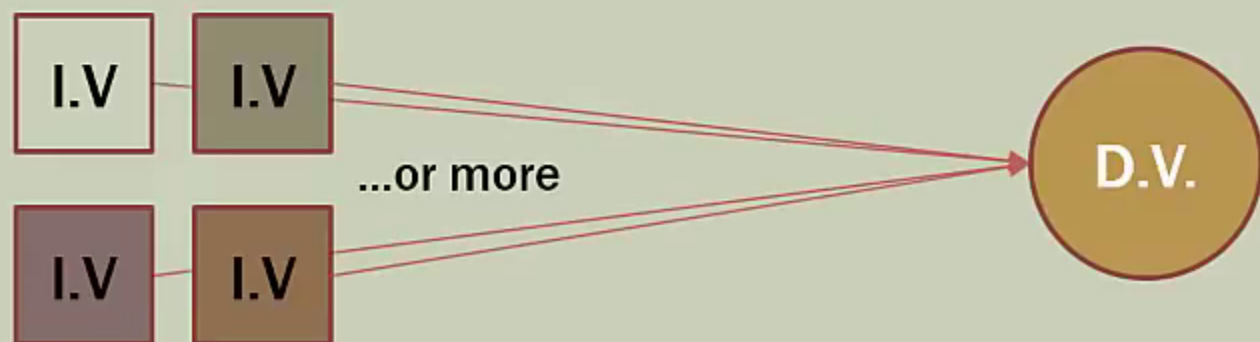
Multiple regression  
*many-to-one*

# MULTIPLE REGRESSION

Multiple regression is an extension of simple linear regression.



Simple linear regression  
*one-to-one*



Multiple regression  
*many-to-one*

# MULTIPLE REGRESSION MODEL

Multiple Regression  
Model

$$y = \boxed{\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots \beta_p x_p} + \boxed{\epsilon}$$

linear parameters error

Multiple Regression  
Equation

$$E(y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots \beta_p x_p$$

error term assumed to be zero

Estimated Multiple  
Regression Equation

$$\hat{y} = b_0 + b_1 x_1 + b_2 x_2 + \cdots b_p x_p$$

$b_0, b_1, b_2, \dots b_p$  are the estimates of  $\beta_0, \beta_1, \beta_2, \dots \beta_p$   
 $\hat{y}$  = predicted value of the dependent variable

# Three-Variable Model

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$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon$$

where

$Y$  = Dependent variable

$X_1$  = 1<sup>st</sup> independent variable

$X_2$  = 2<sup>nd</sup> independent variable

The Least-squares normal equations formula for Two Variable Regression Equation

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2$$

---

We can obtain the parameter estimates  $b_0$ ,  $b_1$ , and  $b_2$  by *normalizing* the above regression equation:

$$\begin{aligned}\Sigma Y &= n\beta_0 + \beta_1 \Sigma X_1 + \beta_2 \Sigma X_2 \\ \Sigma X_1 Y &= \beta_0 \Sigma X_1 + \beta_1 \Sigma X_1^2 + \beta_2 \Sigma X_1 X_2 \\ \Sigma X_2 Y &= \beta_0 \Sigma X_2 + \beta_1 \Sigma X_1 X_2 + \beta_2 \Sigma X_2^2\end{aligned}$$

- Step 1

| Miles Travelled (x1) | NumDeliveries (x2) | TravelTime (Y)    | x1.x2 | X1 <sup>2</sup> | X2 <sup>2</sup> | X1.Y | X2.Y |
|----------------------|--------------------|-------------------|-------|-----------------|-----------------|------|------|
| 89                   | 4                  | 7                 |       |                 |                 |      |      |
| 66                   | 1                  | 5.4               |       |                 |                 |      |      |
| 78                   | 3                  | 6.6               |       |                 |                 |      |      |
| 111                  | 6                  | 7.4               |       |                 |                 |      |      |
| 44                   | 1                  | 4.8               |       |                 |                 |      |      |
| 77                   | 3                  | 6.4               |       |                 |                 |      |      |
| 80                   | 3                  | 7                 |       |                 |                 |      |      |
| 66                   | 2                  | 5.6               |       |                 |                 |      |      |
| 109                  | 5                  | 7.3               |       |                 |                 |      |      |
| 76                   | 3                  | 6.4               |       |                 |                 |      |      |
| $\Sigma X1 = 796$    | $\Sigma X2 = 31$   | $\Sigma Y = 63.9$ |       |                 |                 |      |      |

- Step 1

| Miles Travelled (x1) | NumDeliveries (x2) | TravelTime (Y)    | x1.x2                 | X1 <sup>2</sup> | X2 <sup>2</sup> | X1.Y | X2.Y |
|----------------------|--------------------|-------------------|-----------------------|-----------------|-----------------|------|------|
| 89                   | 4                  | 7                 | 356                   |                 |                 |      |      |
| 66                   | 1                  | 5.4               | 66                    |                 |                 |      |      |
| 78                   | 3                  | 6.6               | 234                   |                 |                 |      |      |
| 111                  | 6                  | 7.4               | 666                   |                 |                 |      |      |
| 44                   | 1                  | 4.8               | 44                    |                 |                 |      |      |
| 77                   | 3                  | 6.4               | 231                   |                 |                 |      |      |
| 80                   | 3                  | 7                 | 240                   |                 |                 |      |      |
| 66                   | 2                  | 5.6               | 132                   |                 |                 |      |      |
| 109                  | 5                  | 7.3               | 545                   |                 |                 |      |      |
| 76                   | 3                  | 6.4               | 228                   |                 |                 |      |      |
| $\Sigma X1 = 796$    | $\Sigma X2 = 31$   | $\Sigma Y = 63.9$ | $\Sigma X1.X2 = 2742$ |                 |                 |      |      |

- Step 1

| Miles Travelled (x1) | NumDeliveries (x2) | TravelTime (Y)    | x1.x2                 | X1 <sup>2</sup>       | X2 <sup>2</sup> | X1.Y | X2.Y |
|----------------------|--------------------|-------------------|-----------------------|-----------------------|-----------------|------|------|
| 89                   | 4                  | 7                 | 356                   | 7921                  |                 |      |      |
| 66                   | 1                  | 5.4               | 66                    | 4356                  |                 |      |      |
| 78                   | 3                  | 6.6               | 234                   | 6084                  |                 |      |      |
| 111                  | 6                  | 7.4               | 666                   | 12321                 |                 |      |      |
| 44                   | 1                  | 4.8               | 44                    | 1936                  |                 |      |      |
| 77                   | 3                  | 6.4               | 231                   | 5929                  |                 |      |      |
| 80                   | 3                  | 7                 | 240                   | 6400                  |                 |      |      |
| 66                   | 2                  | 5.6               | 132                   | 4356                  |                 |      |      |
| 109                  | 5                  | 7.3               | 545                   | 11881                 |                 |      |      |
| 76                   | 3                  | 6.4               | 228                   | 5776                  |                 |      |      |
| $\Sigma X1 = 796$    | $\Sigma X2 = 31$   | $\Sigma Y = 63.9$ | $\Sigma X1.X2 = 2742$ | $\Sigma X1^2 = 66960$ |                 |      |      |

- Step 1

| Miles Traveled (x1) | Number of Deliveries (x2) | Travel Time (Y)   | x1.x2                 | X1 <sup>2</sup>       | X2 <sup>2</sup>     | X1.Y | X2.Y |
|---------------------|---------------------------|-------------------|-----------------------|-----------------------|---------------------|------|------|
| 89                  | 4                         | 7                 | 356                   | 7921                  | 16                  |      |      |
| 66                  | 1                         | 5.4               | 66                    | 4356                  | 1                   |      |      |
| 78                  | 3                         | 6.6               | 234                   | 6084                  | 9                   |      |      |
| 111                 | 6                         | 7.4               | 666                   | 12321                 | 36                  |      |      |
| 44                  | 1                         | 4.8               | 44                    | 1936                  | 1                   |      |      |
| 77                  | 3                         | 6.4               | 231                   | 5929                  | 9                   |      |      |
| 80                  | 3                         | 7                 | 240                   | 6400                  | 9                   |      |      |
| 66                  | 2                         | 5.6               | 132                   | 4356                  | 4                   |      |      |
| 109                 | 5                         | 7.3               | 545                   | 11881                 | 25                  |      |      |
| 76                  | 3                         | 6.4               | 228                   | 5776                  | 9                   |      |      |
| $\Sigma X1 = 796$   | $\Sigma X2 = 31$          | $\Sigma Y = 63.9$ | $\Sigma X1.X2 = 2742$ | $\Sigma X1^2 = 66960$ | $\Sigma X2^2 = 119$ |      |      |

- Step 1

| Miles Travelled (x1) | NumDeliveries (x2) | TravelTime (Y)    | x1.x2                 | X1 <sup>2</sup>       | X2 <sup>2</sup>     | X1.Y                   | X2.Y |
|----------------------|--------------------|-------------------|-----------------------|-----------------------|---------------------|------------------------|------|
| 89                   | 4                  | 7                 | 356                   | 7921                  | 16                  | 623                    |      |
| 66                   | 1                  | 5.4               | 66                    | 4356                  | 1                   | 356.4                  |      |
| 78                   | 3                  | 6.6               | 234                   | 6084                  | 9                   | 514.8                  |      |
| 111                  | 6                  | 7.4               | 666                   | 12321                 | 36                  | 821.4                  |      |
| 44                   | 1                  | 4.8               | 44                    | 1936                  | 1                   | 211.2                  |      |
| 77                   | 3                  | 6.4               | 231                   | 5929                  | 9                   | 492.8                  |      |
| 80                   | 3                  | 7                 | 240                   | 6400                  | 9                   | 560                    |      |
| 66                   | 2                  | 5.6               | 132                   | 4356                  | 4                   | 369.6                  |      |
| 109                  | 5                  | 7.3               | 545                   | 11881                 | 25                  | 795.7                  |      |
| 76                   | 3                  | 6.4               | 228                   | 5776                  | 9                   | 486.4                  |      |
| $\Sigma X1 = 796$    | $\Sigma X2 = 31$   | $\Sigma Y = 63.9$ | $\Sigma X1.X2 = 2742$ | $\Sigma X1^2 = 66960$ | $\Sigma X2^2 = 119$ | $\Sigma X1.Y = 5231.3$ |      |

- Step 1

| Miles Travelled (x1) | Number of Deliveries (x2) | Travel Time (Y)   | x1.x2                 | X1 <sup>2</sup>       | X2 <sup>2</sup>     | X1.Y                   | X2.Y                  |
|----------------------|---------------------------|-------------------|-----------------------|-----------------------|---------------------|------------------------|-----------------------|
| 89                   | 4                         | 7                 | 356                   | 7921                  | 16                  | 623                    | 28                    |
| 66                   | 1                         | 5.4               | 66                    | 4356                  | 1                   | 356.4                  | 5.4                   |
| 78                   | 3                         | 6.6               | 234                   | 6084                  | 9                   | 514.8                  | 19.8                  |
| 111                  | 6                         | 7.4               | 666                   | 12321                 | 36                  | 821.4                  | 44.4                  |
| 44                   | 1                         | 4.8               | 44                    | 1936                  | 1                   | 211.2                  | 4.8                   |
| 77                   | 3                         | 6.4               | 231                   | 5929                  | 9                   | 492.8                  | 19.2                  |
| 80                   | 3                         | 7                 | 240                   | 6400                  | 9                   | 560                    | 21                    |
| 66                   | 2                         | 5.6               | 132                   | 4356                  | 4                   | 369.6                  | 11.2                  |
| 109                  | 5                         | 7.3               | 545                   | 11881                 | 25                  | 795.7                  | 36.5                  |
| 76                   | 3                         | 6.4               | 228                   | 5776                  | 9                   | 486.4                  | 19.2                  |
| $\Sigma X1 = 796$    | $\Sigma X2 = 31$          | $\Sigma Y = 63.9$ | $\Sigma X1.X2 = 2742$ | $\Sigma X1^2 = 66960$ | $\Sigma X2^2 = 119$ | $\Sigma X1.Y = 5231.3$ | $\Sigma X2.Y = 209.5$ |

- Step 2 - Substitute

|                    |                   |                   |                         |                        |                      |                         |                        |
|--------------------|-------------------|-------------------|-------------------------|------------------------|----------------------|-------------------------|------------------------|
| $\Sigma X_1 = 796$ | $\Sigma X_2 = 31$ | $\Sigma Y = 63.9$ | $\Sigma X_1 X_2 = 2742$ | $\Sigma X_1^2 = 66960$ | $\Sigma X_2^2 = 119$ | $\Sigma X_1 Y = 5231.3$ | $\Sigma X_2 Y = 209.5$ |
|--------------------|-------------------|-------------------|-------------------------|------------------------|----------------------|-------------------------|------------------------|

IN

$$\begin{aligned}\Sigma Y &= n\beta_0 + \beta_1 \Sigma X_1 + \beta_2 \Sigma X_2 \\ \Sigma X_1 Y &= \beta_0 \Sigma X_1 + \beta_1 \Sigma X_1^2 + \beta_2 \Sigma X_1 X_2 \\ \Sigma X_2 Y &= \beta_0 \Sigma X_2 + \beta_1 \Sigma X_1 X_2 + \beta_2 \Sigma X_2^2\end{aligned}$$

- $63.9 = 10(b_0) + b_1(796) + b_2(31)$
- $5231.3 = b_0(796) + b_1(66960) + b_2(2742)$
- $209.5 = b_0(31) + b_1(2742) + b_2(119)$

- Step 3 Solve the system of linear equation
- Using Gauss Elimination Method

$$\begin{cases} 10x_1 + 796x_2 + 32x_3 = 63.9 \\ 796x_1 + 66960x_2 + 2742x_3 = 5231.3 \\ 31x_1 + 2742x_2 + 119x_3 = 209.5 \end{cases}$$

Rewrite the system in matrix form and solve it by Gaussian Elimination (Gauss-Jordan elimination)

$$\left( \begin{array}{ccc|c} 10 & 796 & 32 & 63.9 \\ 796 & 66960 & 2742 & 5231.3 \\ 31 & 2742 & 119 & 209.5 \end{array} \right)$$

$R_1 / 10 \rightarrow R_1$  (divide the 1 row by 10)

$$\left( \begin{array}{ccc|c} 1 & 79.6 & 3.2 & 6.39 \\ 796 & 66960 & 2742 & 5231.3 \\ 31 & 2742 & 119 & 209.5 \end{array} \right)$$

$R_2 - 796 R_1 \rightarrow R_2$  (multiply 1 row by 796 and subtract it from 2 row);  $R_3 - 31 R_1 \rightarrow R_3$  (multiply 1 row by 31 and subtract it from 3 row)

$$\left( \begin{array}{ccc|c} 1 & 79.6 & 3.2 & 6.39 \\ 0 & 3598.4 & 194.8 & 144.86 \\ 0 & 274.4 & 19.8 & 11.41 \end{array} \right)$$

$R_2 / 3598.4 \rightarrow R_2$  (divide the 2 row by 3598.4)

$$\left( \begin{array}{ccc|c} 1 & 79.6 & 3.2 & 6.39 \\ 0 & 1 & \frac{487}{8996} & \frac{7243}{179920} \\ 0 & 274.4 & 19.8 & 11.41 \end{array} \right)$$

$R_1 - 79.6 R_2 \rightarrow R_1$  (multiply 2 row by 79.6 and subtract it from 1 row);  $R_3 - 274.4 R_2 \rightarrow R_3$  (multiply 2 row by 274.4 and subtract it from 3 row)

$$\left( \begin{array}{ccc|c} 1 & 0 & -\frac{4989}{4498} & \frac{286573}{89960} \\ 0 & 1 & \frac{487}{8996} & \frac{7243}{179920} \\ 0 & 0 & \frac{11122}{2249} & \frac{4088}{11245} \end{array} \right)$$

$$R_3 / \frac{11122}{2249} \rightarrow R_3 \text{ (divide the 3 row by } \frac{11122}{2249}\text{)}$$

$$\left( \begin{array}{ccc|c} 1 & 0 & -\frac{4989}{4498} & \frac{286573}{89960} \\ 0 & 1 & \frac{487}{8996} & \frac{7243}{179920} \\ 0 & 0 & 1 & \frac{2044}{27805} \end{array} \right)$$

$$R_1 + \frac{4989}{4498} R_3 \rightarrow R_1 \text{ (multiply 3 row by } \frac{4989}{4498} \text{ and add it to 1 row); } R_2 - \frac{487}{8996} R_3 \rightarrow R_2 \text{ (multiply 3 row by } \frac{487}{8996} \text{ and subtract it from 2 row)}$$

$$\left( \begin{array}{ccc|c} 1 & 0 & 0 & \frac{726733}{222440} \\ 0 & 1 & 0 & \frac{16139}{444880} \\ 0 & 0 & 1 & \frac{2044}{27805} \end{array} \right)$$

$$\begin{cases} x_1 = \frac{726733}{222440} \\ x_2 = \frac{16139}{444880} \\ x_3 = \frac{2044}{27805} \end{cases}$$

- $b_0 = \frac{726733}{222440} = 3.2671,$
- $b_1 = \frac{16139}{444880} = 0.0363$
- $b_2 = \frac{2044}{27805} = 0.0735$

Step 4.

- Multi Linear Equation

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2$$

- From the solution of unknown variables we get
- $Y = 3.2671 + 0.0363.X_1 + 0.0735.X_2$
- If  $x_1 = 50$  Miles, &  $x_2 = 5$ . Deliveries Find the total travel time?
- $Y = 3.2671 + 0.0363.(50) + 0.0735.(5)$
- $Y = 5.5$

- Generalized Equation for More than 2 Independent Variable

$$\begin{aligned}\Sigma Y &= n\beta_0 + \beta_1 \Sigma X_1 + \beta_2 \Sigma X_2 \\ \Sigma X_1 Y &= \beta_0 \Sigma X_1 + \beta_1 \Sigma X_1^2 + \beta_2 \Sigma X_1 X_2 \\ \Sigma X_2 Y &= \beta_0 \Sigma X_2 + \beta_1 \Sigma X_1 X_2 + \beta_2 \Sigma X_2^2\end{aligned}$$

$$\begin{array}{ccccccc} n\hat{\beta}_0 & +\hat{\beta}_1 \sum_{i=1}^n x_{i1} & +\hat{\beta}_2 \sum_{i=1}^n x_{i2} & +\cdots & +\hat{\beta}_k \sum_{i=1}^n x_{ik} & = & \sum_{i=1}^n y_i \\ \hat{\beta}_0 \sum_{i=1}^n x_{i1} & +\hat{\beta}_1 \sum_{i=1}^n x_{i1}^2 & +\hat{\beta}_2 \sum_{i=1}^n x_{i1}x_{i2} & +\cdots & +\hat{\beta}_k \sum_{i=1}^n x_{i1}x_{ik} & = & \sum_{i=1}^n x_{i1}y_i \\ \vdots & \vdots & \vdots & & \vdots & & \vdots \\ \hat{\beta}_0 \sum_{i=1}^n x_{ik} & +\hat{\beta}_1 \sum_{i=1}^n x_{ik}x_{i1} & +\hat{\beta}_2 \sum_{i=1}^n x_{ik}x_{i2} & +\cdots & +\hat{\beta}_k \sum_{i=1}^n x_{ik}^2 & = & \sum_{i=1}^n x_{ik}y_i \end{array}$$

# Solve

- The clock prices data set contains the selling Price (in pounds Stirling) of 6 antique grandfather clocks in different auctions, along with the Age of the clocks in years and the number of Bidders participating in that auction. Create a regression model for the Price (Dependent Variable) based on Age and Bidders.

| Age | Bidders | Price |
|-----|---------|-------|
| 127 | 13      | 1235  |
| 115 | 12      | 1080  |
| 127 | 7       | 845   |
| 150 | 9       | 1522  |
| 156 | 6       | 1047  |
| 182 | 11      | 1979  |

| Age<br>X1 | Bidders<br>X2 | Price<br>Y | x1.x2 | X1 <sup>2</sup> | X2 <sup>2</sup> | X1.Y    | X2.Y  |
|-----------|---------------|------------|-------|-----------------|-----------------|---------|-------|
| 127       | 13            | 1235       | 1651  | 16129           | 169             | 156845  | 16055 |
| 115       | 12            | 1080       | 1380  | 13225           | 144             | 124200  | 12960 |
| 127       | 7             | 845        | 889   | 16129           | 49              | 107315  | 5915  |
| 150       | 9             | 1522       | 1350  | 22500           | 81              | 228300  | 13698 |
| 156       | 6             | 1047       | 936   | 24336           | 36              | 163332  | 6282  |
| 182       | 11            | 1979       | 2002  | 33124           | 121             | 360178  | 21769 |
| 857       | 58            | 7708       | 8208  | 125443          | 600             | 1140170 | 76679 |

- Substituting in Least Square Formula
- $7708 = b_0 + b_1 \cdot 0.6 + b_2 \cdot 58$
- $1140170 = b_0 + b_1 \cdot 0.857 + b_2 \cdot 1.125443$
- $76679 = b_0 + b_1 \cdot 0.58 + b_2 \cdot 1.8208$
- Solving the above equation using Gauss Elimination Method
- $b_0 = -1678.7483$ ,  $b_1 = 15.0409$ ,  $b_2 = 84.3168$
- Regression Line Equation
- $Y = -1678.7483 + 15.0409 \cdot x_1 + 84.3168 \cdot x_2$
- If  $x_1 = 170$  &  $x_2 = 10$ , the estimated bid will be
- $Y = -1678.7483 + 15.0409 \cdot (170) + 84.3168 \cdot (10)$
- $Y = 1721.3727$